

Analysis of Sequence Variation

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Sequence Alignment

Sequence Alignment

- ▶ Provides a measure of relatedness
- ▶ Alignment quantified by similarity (% identity)
- ▶ Useful for any sequential data type:
 - ▶ DNA/RNA
 - ▶ Amino acids
 - ▶ Protein secondary structure
- ▶ High sequence similarity might imply:
 - ▶ Common evolutionary history
- ▶ Similar biological function

What Alignments Can Tell Us

- ▶ Homology - Orthologs, Paralog
- ▶ Genomic identity/origin of a sequence/individual
- ▶ Genome/gene structure
- ▶ Genic structure (exons, introns, etc)
 - ▶ RNA 2D structure
 - ▶ Chromosome rearrangements/3D structure

DNA Sequence Alignment Example

Sequence 1 ATACACAGTAGGAGATACCAGTAAGGGAGGGGG

Sequence 2 ATACCATAAGCGAG

Alignment 1

	Match	Mismatch
ATACACAGTAGGAGATACCAGTAAGGGAGGGGG		
-----ATACCA-TAAGCGAG----		

Gap

Alignment 2

ATACACAGTAGGAGATACCAGTAAGGGAGGGGG
ATAC-CA-----TAAGCGAG----

Alignment 3

ATACACAGTAGGAGATACCAGTAAGGGAGGGGG
ATAC-CA-TA--AG---C--G--AG-----

Scoring/Substitution Matrices

- ▶ Given alignment, how “good” is it?
- ▶ Higher score = better alignment
- ▶ Implicitly represent evolutionary patterns

	A	C	G	T	-
A	2	-3	-1	-3	-3
C	-3	2	-3	-1	-3
G	-1	-3	2	-3	-3
T	-3	-1	-3	2	-3
-	-3	-3	-3	-3	NA

ATACCAG**G**TAAG**G**GAG
 ATACCA-TAAG**A**GAG

Score = 22

ATACCAG**G**TAAG**G**-GAG
 ATACCA-TAAG-**A**GAG

Score = 19

ATACCA-**G**TAAG**G**GAG
 A-TACCATAAG**A**GAG-

Score = -20

Sequence Alignment Algorithms

- ▶ **Global** alignments - beginning and end of both sequences must align
- ▶ **Local** alignments - one sequence may align anywhere within the other
- ▶ Multiplicity:
 - ▶ Pairwise alignments (2 sequences)
 - ▶ Multiple sequence alignment (3+ sequences)

Global Alignment

Both sequences are aligned from end to end

```
AAANTAIYYDPNPDMP A--
      NTAI-YDPN--M-
```

Interior sequences are aligned as well as possible

```
AERAKDNLCRLEHTTLRKVTAAANTAIYYDPNPDMPVVAEDQEWWNVYYEM
A-----N-----T-----AI-YD--P-----N----M
```

However, sequences of vastly different length can produce meaningless alignments

Local Alignment

Alignment may begin and end at any position

```
AAANTAIYYDPNPDMP -
AANTAI-YDPN--M-
```

```
AERAKDNLCRLEHTTLRKVTAAANTAIYYDPNPDMPVVAEDQEWVNVYYEM
-----AANTAI-YDPN--M-----
```

Local alignment may produce better alignments when
sequence lengths differ greatly

Multiple Sequence Alignment

Like pairwise alignment, but with N sequences

```

Q5E940_BOVIN -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_HUMAN -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_MOUSE -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_RAT -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_CHICK -----MPREDRATKSNFYFKIIQLDDYPKCFVVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_RANSY -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--SALE
Q7ZUG3_BRARE -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_ICTFU -----MPREDRATKSNFYFKIIQLDDYPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_DROME -----MYRENKAANKAQYFKVYELFDEFPKCFIVGADNVGSKMQQIEMSLRGK-AVVLGKNTMMRKAIRGHLENN--PALE
RLA0_DICDI -----MSGAG-SKRKNVFEKATKLFITYDKHIVAEADFFGSSQLKIRKSIIRGI-GAVLMGKNTMIRKVIINDLADSK--PELD
Q54LP0_DICDI -----MSGAG-SKRKNVFEKATKLFITYDKHIVAEADFFGSSQLKIRKSIIRGI-GAVLMGKNTMIRKVIINDLADSK--PELD
RLA0_PLAF8 -----MAKLSQKKQKQMYIEKLSLIQYYSKILIVHVDNVGSKMASVYKSLRGK-ATILMGKNTIRITALKKNLQAV--POIE
RLA0_SULAC -----HIGLAVTTTKKIAKKYDEVAELTKLTKTIIIANIEGFPADKLHEIKKLRGK-ADIKVTKNLFNIAKKNAG----GYEK
RLA0_SULTO -----MRIMAVITQERKIAKKIEEVKELQKREYHTIIIANIEGFPADKLHEIKKLRGK-ADIKVTKNLFNIAKKNAG----LDYS
RLA0_SULSO -----MKRLALALKQRKVASWKEELVKEELIKNSHTILIGNIEGFPADKLHEIKKLRGK-ADIKVTKNLFNIAKKNAG----IDIE
RLA0_AERPE MSVVSIVGOMYKREKIPDEWKTLMRELELFSEKRVVLFADLTCTPTFFVQVYKCKLWKK--YPMVAKKRIILSAMKAAGLE---LDDN
RLA0_PYRAE HMLAIGKRRYVRTQYVARKVKIVSEATLLQKQYVYVFLFDLHGLSERILHEVYRLERY-GVIRIKIPFLFKIAFTKYVG---IPAE
RLA0_METAC -----MAERHHTENIPQKKDEIEMIKELIQSHKVFQMVGIEGILATKHKIRDLKDV-AVLKVRNTLTERALNQLG---ETIP
RLA0_METMA -----MAERHHTENIPQKKDEIEMIKELIQSHKVFQMVRIEGILATKHKIRDLKDV-AVLKVRNTLTERALNQLG---ESIP
RLA0_ARCFU -----MAAVRGS-----PPEYKRAVEEIKRMISSEKPVVAIVSFRNVPAGQMKIRREFRGK-AEIKVYKNTLLERDALDGL---GOYL
RLA0_METKA MAVKAKGQPSQYEKPKVAEKRRYVKEKLMDENYGVLDVLEIPAPOLQETRAKLERDEIIRMSKNTLMRTALEKKLDER--PELE
RLA0_METHH -----MAHVAEKKKEVQQLHDLIKSYVVGIANLADIPARQLKMQQLADS-ALIRMSKNTLISLAEKAGREL--ENVD
RLA0_METTL -----MITAESENKIAVKKIEEVNKKLLKNGQIVAVDMMVFPARQLQETRAKIR-DMTLMKSRNTLIRAEKVAEATQNPFA
RLA0_METVA -----MIDAKSENKIAVKKIEEVNKKLLKSNVIALIDHMEVPAYQLQETRAKIR-DQMTLMKSRNTLIRAEKVAEATQNPFA
RLA0_METJA -----METVKANVAPKIEEVNKKLLKSLKSVVAIVDMMVFPARQLQETRAKIR-DKVKLMKSRNTLIRAEKVAEATQNPFA
  
```



Sequence consensus among many species suggests evolutionary pressure

Methods for Multiple Sequence Alignment (MSA)

1. Progressive Alignment Algorithms:

- ▶ *ClustalW*: A widely used progressive alignment tool with a guide tree strategy.
- ▶ *Clustal Omega*: An enhanced version of ClustalW with improved speed and accuracy.

2. Iterative Alignment Algorithms:

- ▶ *MAFFT (Multiple Alignment using Fast Fourier Transform)*: Uses iterative refinement with consistency scores.
- ▶ *MUSCLE (Multiple Sequence Comparison by Log-Expectation)*: Utilizes progressive alignment followed by iterative refinement.

Methods for MSA (Continued)

3. Hidden Markov Models (HMMs):

- ▶ *HMMER*: Based on HMMs, used for alignment and homology detection.
- ▶ *SAM (Sequence Alignment and Modeling System)*: Combines HMMs with profiles for database searches.

4. Probabilistic Alignment Methods:

- ▶ *ProbCons*: Generates a probabilistic alignment using a Bayesian framework.
- ▶ *PRANK*: Considers sequence and alignment uncertainty in alignment generation.

Methods for MSA (Continued)

5. Structure-Based Alignment:

- ▶ *MUSTANG (Multiple Structural Alignment by Secondary Structures)*: Aligns based on protein structures considering sequence and structure.
- ▶ *DALI (Distance Alignment Matrix Method)*: Aligns sequences based on structural similarity.

These methods vary in their approaches and are chosen based on factors such as alignment accuracy, computational efficiency, and the characteristics of the input sequences.

ClustalW: A Common MSA Tool

- ▶ ClustalW is one of the most widely used tools for multiple sequence alignment.
- ▶ It uses a progressive alignment approach.
- ▶ Available as standalone software or through a web server.

Example: Aligning TB genomes

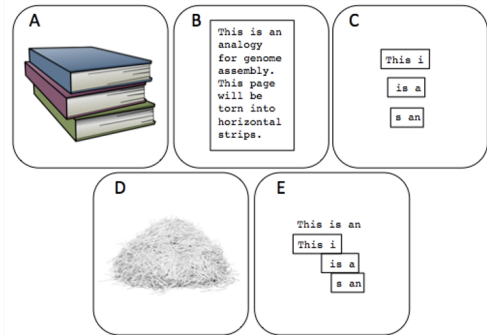
Download the following TB genomes:

- ▶ H37Rv (Click here to download)
- ▶ Mycobacterium tuberculosis str. Erdman (Click here)
- ▶ Save the fasta data/sequence into separate files
- ▶ Combine into single FASTA, first 100 lines:

```
{ head -101 sequence.fasta; head -101 sequence-2.fasta; } \  
> combined.fasta
```

- ▶ Analyze using ClustalW

Example: Genome Assembly



If your genome was a book that had its sentences chopped into fragments, assembly is analogous to reconstructing all the sentences.

We need multiple copies of each book (genome) to arrive at a *consensus* text (DNA sequence) of the original

Multiple Copies of a Genome

Reads

High Coverage

Low Coverage

Consensus Sequence

Example: Genome Assembly

	ATGGCATTGCAA	
	TGGCATTGCAATTTG	
Reads	AGATGGTATTG	An error? A polymorphism? A different allele? Incorrect alignment?
	GATGGCATTGCAA	
	GCATTGCAATTTGAC	
	ATGGCATTGCAATTT	
	AGATGGTATTGCAATTTG	
Consensus Sequence	AGATGGCATTGCAATTTGAC	

Greedy approach: take most frequent nucleotide at each aligned position

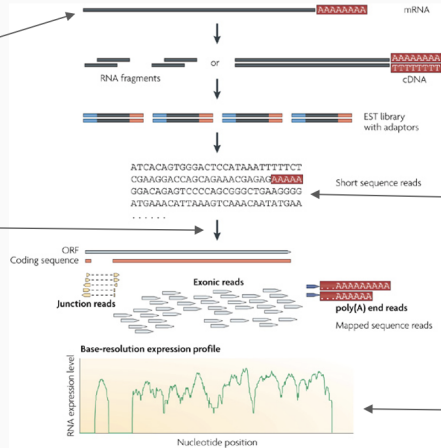
Example: mRNA-Seq Analysis

Start with a pool of mRNA molecules

Find all locations where sequences map in genome

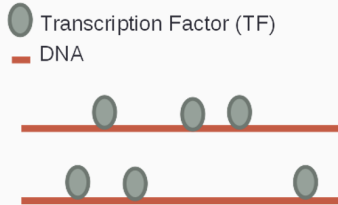
Millions of DNA sequences 30-150 nucleotides long

Count the number of sequences that map to individual regions (e.g. genes)



Example: DNA Binding Site Discovery

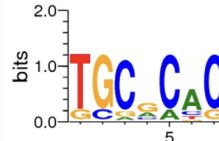
Identify genomic regions where a particular TF is bound across the entire genome



By extracting and aligning the DNA sequence corresponding to these binding events, we can identify which DNA sequences this TF tends to bind

```

. . . T G C T C A C . . .
. . . T G C A C A C . . .
. . . G G C G C A C . . .
. . . T G A G C A C . . .
. . . T C G C C T C . . .
. . . T G C A A C G . . .
. . . T C C C A C G . . .
. . . G G T A C T C . . .
  
```



Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.5         tools_4.5.2
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.17.1 yaml_2.3.12
## [9] rmarkdown_2.30    knitr_1.51        xfun_0.55         digest_0.6.39
## [13] rlang_1.1.6       evaluate_1.0.5
```